



Killara
HIGH SCHOOL

MATHEMATICS DEPARTMENT

Student Number

2024 Year **12** Examination

Mathematics **Advanced**

Task **2**

Date: 12/03/2024

General

Instructions

- Reading time – **5** minutes
- Working time – **55** minutes
- Write using blue or black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- Show relevant mathematical reasoning and/or calculations
- No white-out may be used

Total Marks:

41

Section I - **5** marks

- Allow about **5** minutes for this section

Section II - **36** marks

- Allow about **50** minutes for this section

Q	Marks
MC	/5
6	/6
7	/2
8	/2
9	/3
10	/3
11	/2
12	/4
13	/4
14	/5
15	/5
Total	/41

This question paper must not be removed from the examination room.

*This assessment task constitutes **20%** of the course.*

Section I

5 marks

Attempt Questions 1–5

Allow about 5 minutes for this section.

Use the multiple-choice sheet for Question 1–5.

1 Find the derivative of $f(x) = e^{2x}$ with respect to x .

A. $\frac{d}{dx}f(x) = \frac{1}{2}e^x$

B. $\frac{d}{dx}f(x) = \frac{1}{2}e^{2x}$

C. $\frac{d}{dx}f(x) = 2e^x$

D. $\frac{d}{dx}f(x) = 2e^{2x}$

2 The vertical and horizontal asymptotes of the following function

$$f(x) = \frac{x - 3}{2 - x}$$

are

A. $x = 2, \quad y = -1$

B. $x = 2, \quad y = 1$

C. $x \neq 2, \quad y = -\frac{3}{2}$

D. $x \neq 2, \quad y = \frac{3}{2}$

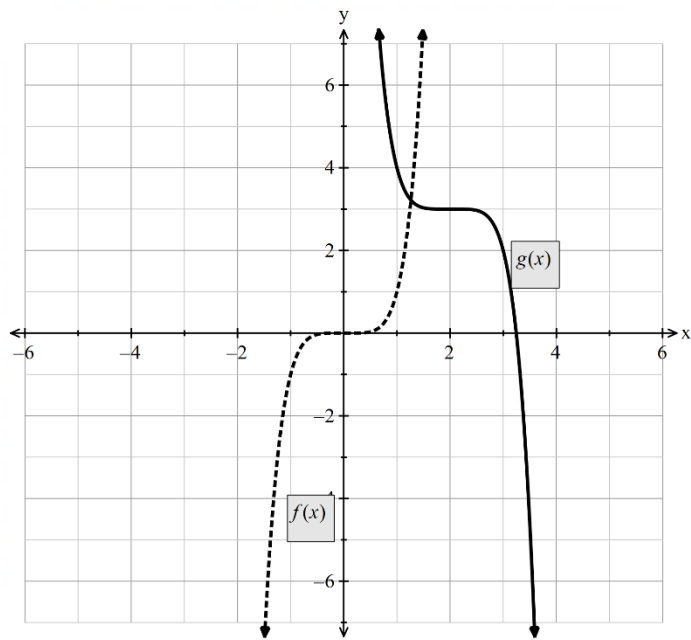
3 What are the changes in amplitude and period when the function $f(x) = 2 \cos \frac{x}{2}$ is transformed into $g(x) = \cos \frac{x}{4}$?

- A. The amplitude is halved and the period is halved
- B. The amplitude is halved and the period is doubled
- C. The amplitude is doubled and the period is halved
- D. The amplitude is doubled and the period is doubled

4 Differentiate $y = \frac{\tan x}{x}$ with respect to x .

- A. $\frac{dy}{dx} = \frac{\sec x^2 - \tan x}{x^2}$
- B. $\frac{dy}{dx} = \frac{\tan x - \sec^2 x}{x^2}$
- C. $\frac{dy}{dx} = \frac{x \sec^2 x - \tan x}{x^2}$
- D. $\frac{dy}{dx} = \frac{\tan x - x \sec^2 x}{x^2}$

- 5 What are the transformations from $f(x)$ to $g(x)$?



- A. Shift to right by 2 units, shift up by 3 units, reflect on x-axis
- B. Shift to left by 2 units, shift up by 3 units, reflect on x-axis
- C. Shift to right by 2 units, shift up by 3 units, reflect on y-axis
- D. Shift to left by 2 units, shift up by 3 units, reflect on y-axis

End of Section I

Section II

36 marks

Attempt Questions 6–15

Allow about 50 minutes for this section.

In Questions 6 – 15, your response should include relevant mathematical reasoning and/or calculations.

Question 6 (6 marks)

Differentiate the following with respect to x :

(a) $y = \frac{1}{3}(x + 4)^6$ 2

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(b) $y = \log_e(2x^2 + 3)$ 2

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(c) $f(x) = \frac{\sin(3x - 1)}{\cos x}$ 2

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Question 7 (2 marks)

Solve $\sin(2x) = -\frac{\sqrt{3}}{2}$, for $-\pi \leq x \leq \pi$.

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Question 8 (2 marks)

If $y = xe^{2x}$, prove that $\frac{dy}{dx} - 2y = e^{2x}$.

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Question 9 (3 marks)

Find the x -values for the points on the graph of $y = \tan x$ at which the normal has slope of $-\frac{1}{2}$, where $0 \leq x \leq \pi$.

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Question 10 (3 marks)

Justin invested \$40 000 into an account with an interest rate of 4.5% compounded per annum. After t years, his account's worth is modelled by

$$A(t) = 40000e^{0.045t}$$

Where A is the total investment at the end of t years.

- (a) Find the total investment after 10 years.

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- (b) Find the rate of change in the value of the account at the end of 10 years. Answer to 2 decimal places.

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Question 11 (2 marks)

The following transformations are applied to the function $y = \ln(x)$:

2

- i) Reflected on x -axis
- ii) Vertical dilation of factor $\frac{1}{2}$
- iii) Horizontal translation to the left by 2 units
- iv) Vertical translation down by 3 units.

State the transformed function.

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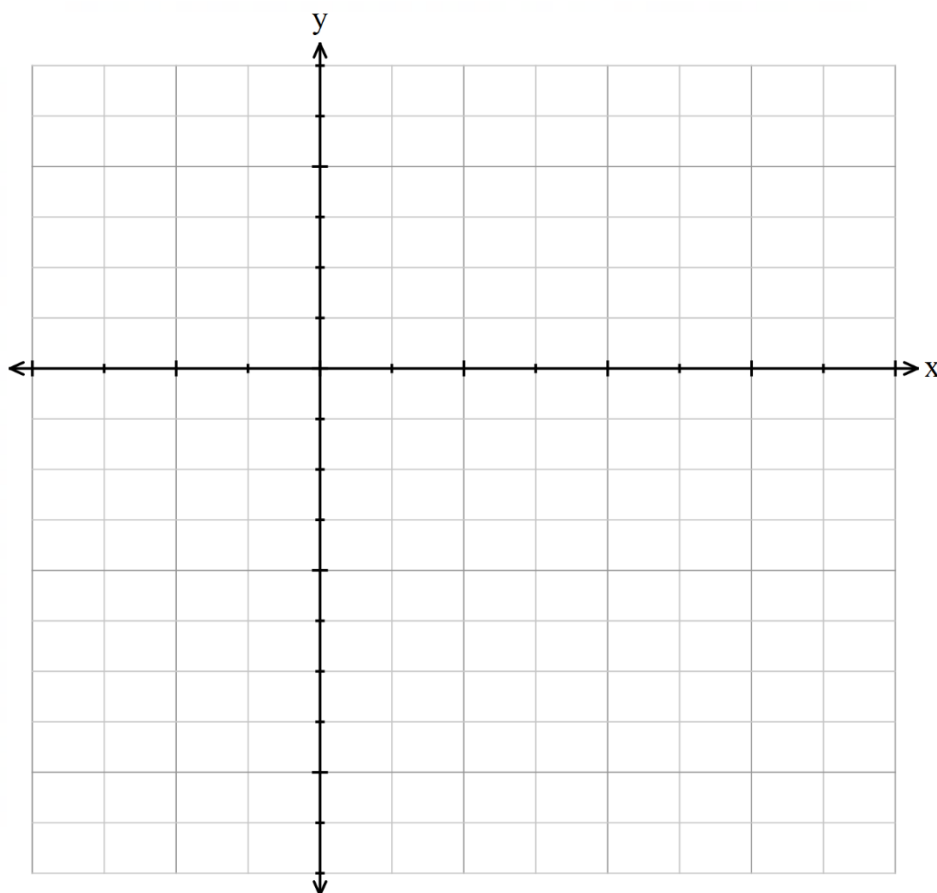
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Question 12 (4 marks)

Given two functions of $f(x) = \ln(x + 1)$ and $g(x) = (x - 1)^2 - 4$.

- (a) Graph $f(x)$ and $g(x)$ on the same graphing template given below, label any key features which may include any asymptotes, x – and y – intercepts and vertex. **3**



- (b) Hence find the number of solutions for the equation $\ln(x + 1) = (x - 1)^2 - 4$. **1**

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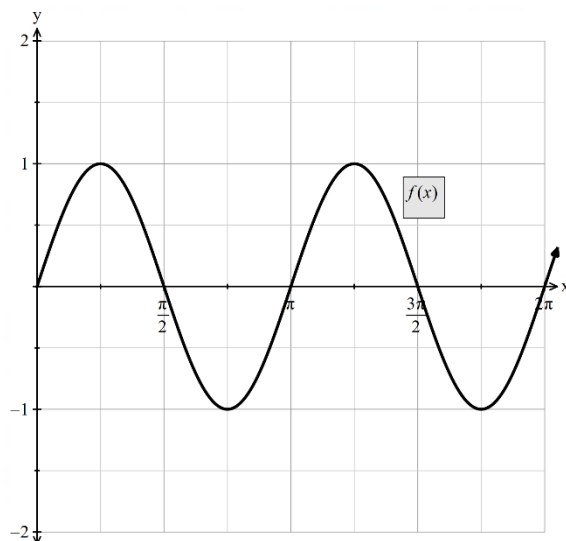
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Question 13 (4 marks)

Given the graph $f(x) = \sin(ax)$ as shown below:



- (a) Determine the value of a , the coefficient of x .

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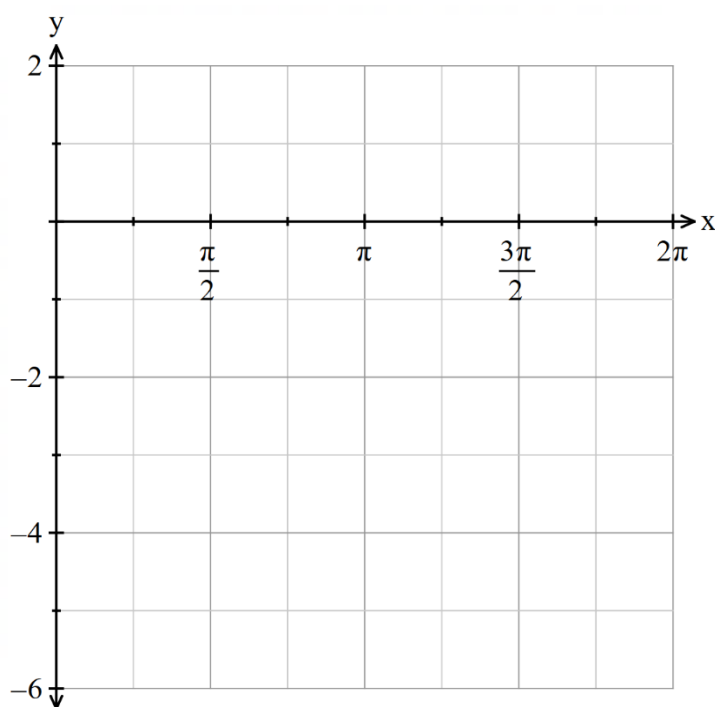
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- (b) The function $f(x) = \sin(ax)$ is then transformed to the function

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$$g(x) = 2 \sin\left(a\left(x + \frac{\pi}{4}\right)\right) - 3.$$

Sketch $g(x)$ on the graph template below. labelling minimum and maximum points.



Question 14 (5 marks)

Tom is jumping on a trampoline. He starts at a minimum height of 30cm below the trampoline frame. Tom can reach a maximum height of 50cm above the trampoline frame.

The model function is

$$H(t) = a \sin(2\pi(t - c)) + 10,$$

where H is the height in cm above the trampoline frame and t is the time in seconds.

- (a) Explain why the amplitude of the function, $a = 40$.

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- (b) Show that $c = \frac{1}{4}$, if $0 < c < 1$.

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- (c) At what time does Tom first reach a height of 30cm above the trampoline? Answer to 2 decimal places.

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Question 15 (5 marks)

Given two functions $f(x) = \frac{-x + 2}{x^2 - 4}$ and $g(x) = \frac{-x + 3}{(x - 1)^2 - 4} - 1$.

- (a) Describe the transformation from $f(x)$ to $g(x)$.

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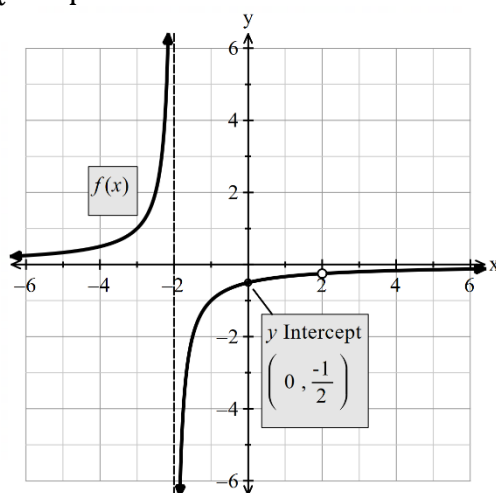
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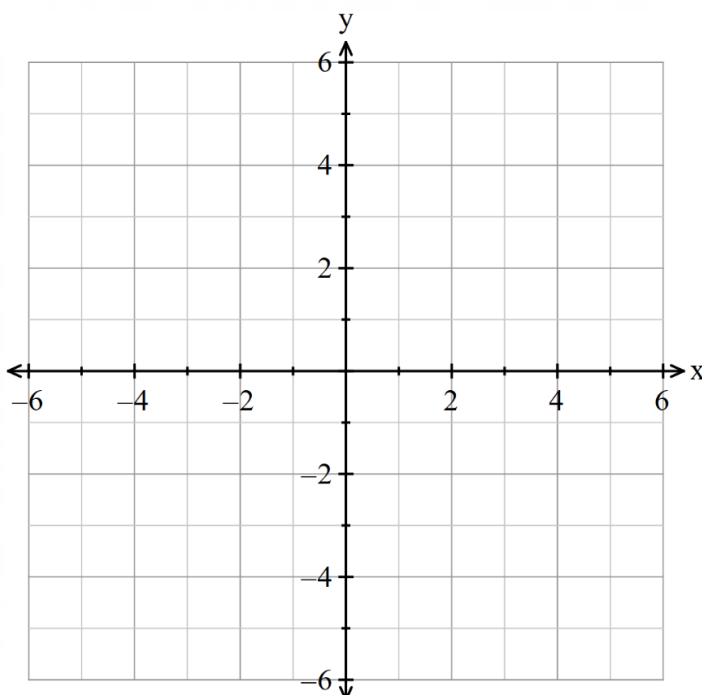
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- (b) The graph of $f(x) = \frac{-x + 2}{x^2 - 4}$ as shown below.

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Sketch $g(x) = \frac{-x + 3}{(x - 1)^2 - 4} - 1$. Label key features.



End of paper

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☒ D. $\frac{d}{dx}f(x) = 2e^{2x}$

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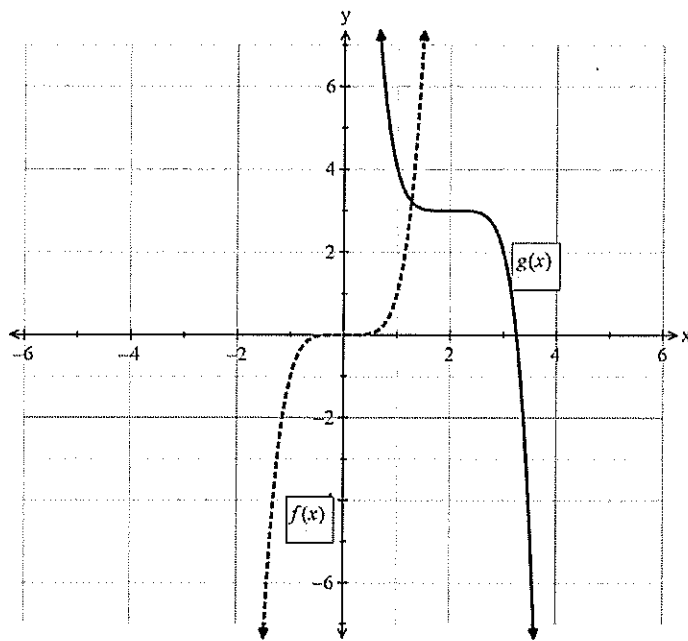
A. $\frac{dy}{dx} = \frac{\sec x^2 - \tan x}{x^2}$

B. $\frac{dy}{dx} = \frac{\tan x - \sec^2 x}{x^2}$

☒ C. $\frac{dy}{dx} = \frac{x \sec^2 x - \tan x}{x^2}$

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- B. Shift to left by 2 units, shift up by 3 units, reflect on x-axis
- C. Shift to right by 2 units, shift up by 3 units, reflect on y-axis
- ☒ D. Shift to left by 2 units, shift up by 3 units, reflect on y-axis

End of Section I

Section II

42 marks

Attempt Questions 6–15

Allow about 50 minutes for this section.

In Questions 6 – 15, your response should include relevant mathematical reasoning and/or calculations.

Question 6 (6 marks)

Differentiate the following with respect to x :

(a) $y = \frac{1}{3}(x+4)^6$ 2

$$\frac{dy}{dx} = \frac{6}{3}(x+4)^5$$
$$= 2(x+4)^5$$

(b) $y = \log_e(2x^2 + 3)$ 2

$$y = \ln(2x^2 + 3)$$
$$\frac{dy}{dx} = \frac{4}{2x^2 + 3}$$

(c) $f(x) = \frac{\sin(3x-1)}{\cos x}$ 2

$$f'(x) = \frac{3\cos(3x-1)\cos x - (\sin 3x - 1)\sin x}{\cos^2 x}$$
$$= \frac{3\cos(3x-1)\cos x + (\sin 3x - 1)\sin x}{\cos^2 x}$$

Question 7 (2 marks)

Solve $\sin(2x) = -\frac{\sqrt{3}}{2}$, for $-\pi \leq x \leq \pi$.

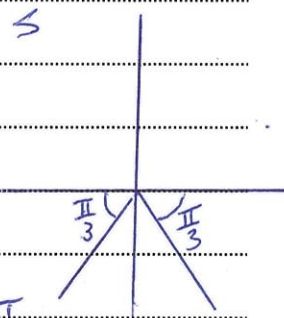
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$$2x = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$2x = \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$x = \frac{2\pi}{3}, \frac{5\pi}{6}$$

$$x = \frac{2\pi}{3}, \frac{5\pi}{6}, -\frac{2\pi}{3}, -\frac{5\pi}{6}$$



Question 8 (2 marks)

If $y = xe^{2x}$, prove that $\frac{dy}{dx} - 2y = e^{2x}$.

2

$$\frac{dy}{dx} = e^{2x} + 2xe^{2x}$$

Since $y = xe^{2x}$

$$\therefore \frac{dy}{dx} = e^{2x} + 2y$$

$$\frac{dy}{dx} - 2y = e^{2x}$$

Question 9 (3 marks)

Find the x -value for the points on the graph of $y = \tan x$ at which the normal has

3

slope of $-\frac{1}{2}$, where $0 \leq x \leq \pi$.

$$\text{slope of normal} = -\frac{1}{2} \quad \therefore y' = 2$$

$$y = \tan x$$

$$y' = \sec^2 x$$

$$2 = \frac{1}{\cos^2 x}$$

$$\cos^2 x = \frac{1}{2}$$

$$\cos x = \pm \frac{\sqrt{2}}{2}$$

$$\therefore x = \frac{\pi}{4}, \text{ or } \pi - \frac{\pi}{4}$$

$$x = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

Question 10 (3 marks)

Justin invested \$40 000 into an account with an interest rate of 4.5% compounded per annum. After t years, his account's worth is modelled by

$$A(t) = 40000e^{0.045t}$$

Where A is the total investment at the end of t years.

- (a) Find the total investment after 10 years.

1

$$A(t) = 40000e^{0.045t}$$

$$A(10) = 40000e^{0.045(10)}$$

$$A(10) = 62732.4874$$

$$\approx \$62732.50$$

- (b) Find the rate of change in the value of the account at the end of 10 years. Answer to 2 decimal places.

2

$$A'(t) = 40000[e^{0.045t} \cdot (0.045)]$$

$$A'(t) = 1800e^{0.045t}$$

$$A'(10) = 1800e^{0.045(10)}$$

$$\approx \$2822.96/\text{year.}$$

Account is increasing at a rate of \$2822.96/year at the end of 10 years.

Question 11 (2 marks)

The following transformations are applied to the function $y = \ln(x)$:

2

- i) Reflected on x -axis
- ii) Vertical dilation of factor $\frac{1}{2}$
- iii) Horizontal translation to the left by 2 units
- iv) Vertical translation down by 3 units.

State the transformed function.

$$(i) \quad y = -\ln x$$

$$(ii) \quad 2y = -\ln x \\ y = -\frac{1}{2} \ln x$$

$$(iii) \quad 2y = -\ln(x+2) \\ y = -\frac{1}{2} \ln(x+2)$$

$$(iv) \quad y = -\frac{1}{2} \ln(x+2) - 3$$

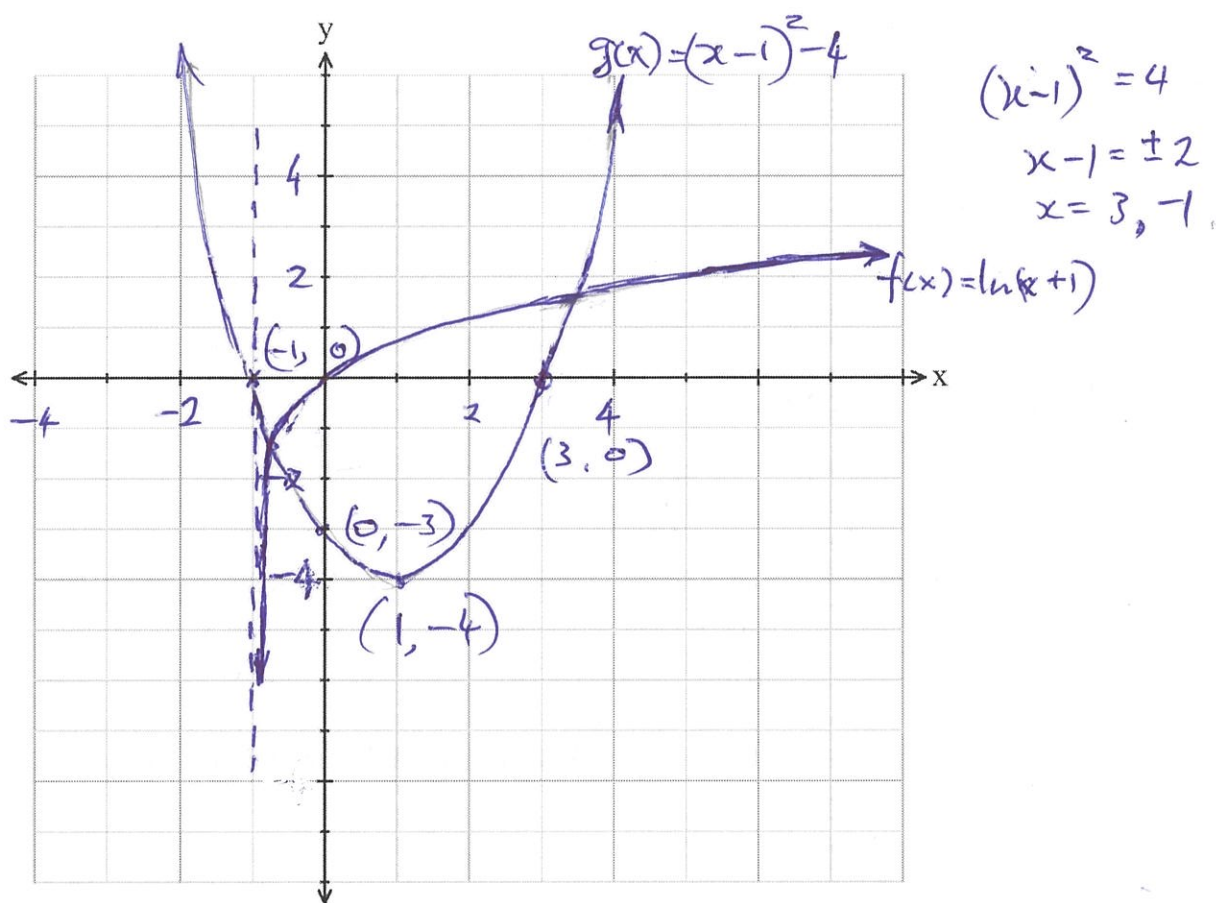
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Question 12 (4 marks)

Given two functions of $f(x) = \ln(x + 1)$ and $g(x) = (x - 1)^2 - 4$.

- (a) Graph $f(x)$ and $g(x)$ on the same graphing template given below, labelling key features of both graphs.

3



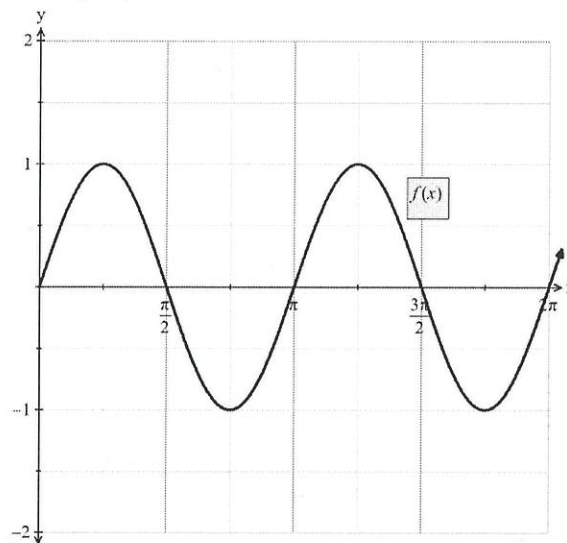
- (b) Hence find the number of solutions for the equation $\ln(x + 1) = (x - 1)^2 - 4$.

1

2 solutions.

Question 13 (4 marks)

Given the graph $f(x) = \sin(ax)$ as shown below:



- (a) Determine a

$$\frac{2\pi}{2} = \pi$$

$$a = 2$$

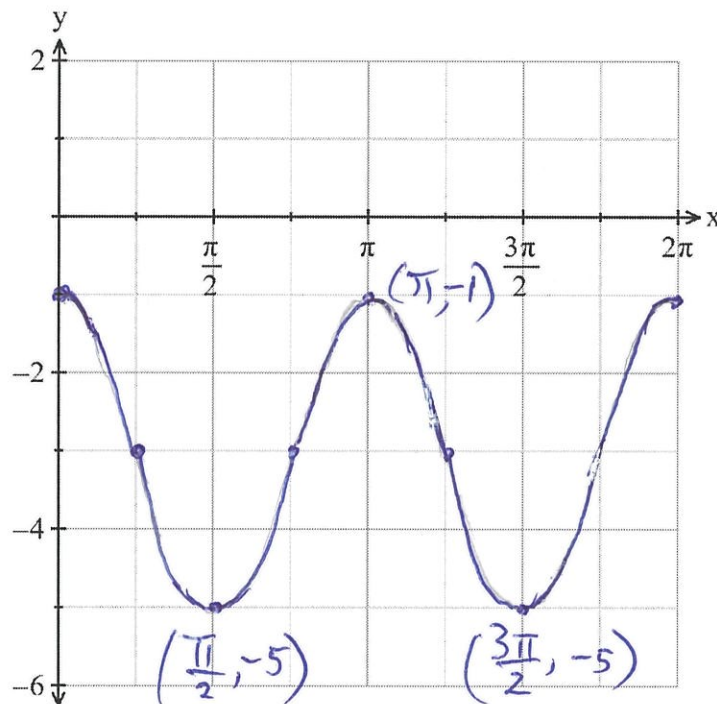
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- (b) The function $f(x) = \sin(ax)$ is then transformed to the function

$$g(x) = 2 \sin\left(a\left(x + \frac{\pi}{4}\right)\right) - 3.$$

3

Sketch $g(x)$ on the graph template below, labelling minimum and maximum points.



Question 14 (5 marks)

Given two functions $f(x) = \frac{-x+2}{x^2-4}$ and $g(x) = \frac{-x+3}{(x-1)^2-4} - 1$.

- (a) Describe the transformation from $f(x)$ to $g(x)$.

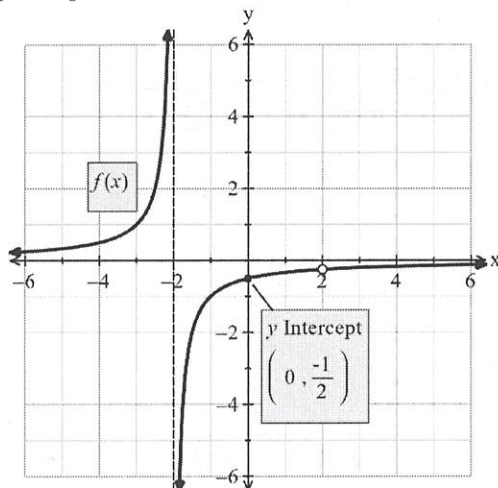
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$$g(x) = \frac{-(x-1)+2}{(x-1)^2-4} - 1$$

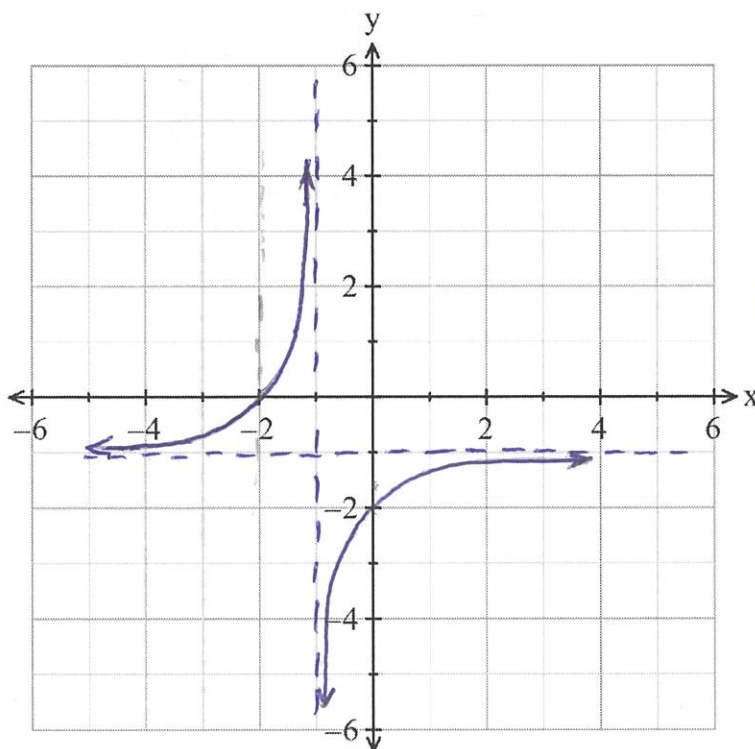
Horizontal translate to right by 1 unit
then vertical translate down by 1 unit.

- (b) The graph of $f(x) = \frac{-x+2}{x^2-4}$ as shown below.

3



Sketch $g(x) = \frac{-x+3}{(x-1)^2-4} - 1$. Label key features.



x -intercept is $x = -2$
 y -intercept is $y = -2$

Question 15 (5 marks)

Tom is jumping on a trampoline. He starts at a minimum height of 30cm below the trampoline frame. Tom can reach a maximum height of 50cm above the trampoline frame.

The model function is

$$H(t) = a \sin(2\pi(t - c)) + 10,$$

where H is the height in cm above the trampoline frame and t is the time in seconds.

- (a) Explain why the amplitude of the function, $a = 40$.

1

$$\begin{aligned} \text{Min height} &= -30 \\ \text{max height} &= 50 \\ \text{Amplitude} &= \frac{50 - (-30)}{2} \\ &= 40 \text{ cm} \end{aligned}$$

- (b) Show that $c = \frac{1}{4}$, if $0 < c < 1$.

2

$$\begin{aligned} \text{when } t=0, \quad 40 \sin(2\pi \times (-c)) + 10 &= -30 \\ \sin(2\pi \times (-c)) &= \frac{-40}{40} \\ 2\pi \times (-c) &= \sin^{-1}(-1) \\ -2\pi c &= -\frac{1}{2}\pi \\ c &= -\frac{1}{2} \times -\frac{1}{2} \\ c &= \frac{1}{4} \end{aligned}$$

- (d) At what time does Tom first reach a height of 30cm above the trampoline? Answer to 2 decimal places.

$$30\text{cm} = 40\text{cm} \sin(2\pi(t - \frac{1}{4})) + 10$$

$$\frac{30-10}{40} = \sin(2\pi(t - \frac{1}{4}))$$

$$\frac{1}{2} = \sin(2\pi t - \frac{2\pi}{4})$$

$$\frac{1}{2} = \sin(2\pi t - \frac{\pi}{2})$$

$$2\pi t - \frac{\pi}{2} = \frac{\pi}{6}$$

$$2\pi t = \frac{\pi}{6} + \frac{\pi}{2}$$

$$2\pi t = \frac{2\pi}{3}$$

$$t = \frac{1}{3}$$

End of paper